STA314: Statistical Methods for Machine Learning I

Midterm Exam – LEC0101
Total time: 2 hours. Total scores: 40 points.

Problem 1 (10 pts) Consider the Lasso estimator which solves the following problem

$$\min_{\beta_1,\beta_2} \sum_{i=1}^{n} \left(y_i - \beta_1 x_{i1} - \beta_2 x_{i2}\right)^2, \quad \text{subject to} \quad |\beta_1| + |\beta_2| \leq s. \tag{1}$$

(1) (1 pt) Does there exist a range of choices of s such that the lasso estimator in Eq. (1) is sparse? Briefly explain.

(2) (1 pt) Does there exist a range of choices of s such that the lasso estimator in Eq. (1) is generally non-sparse? Briefly explain.

(3) (3 pts) Let $\widehat{\boldsymbol{\beta}}^s$ be the solution to Eq. (1) with parameter s. For $\widehat{\boldsymbol{\beta}}^{s_1}$ and $\widehat{\boldsymbol{\beta}}^{s_2}$ with $s_1 \leq s_2$, which corresponding predictor has smaller training MSE? Also compare with the OLS estimator.

(4) (3 pts) Compare the test MSE of the predictors corresponding to $\widehat{\boldsymbol{\beta}}^{s_1}$, $\widehat{\boldsymbol{\beta}}^{s_2}$, and the OLS estimator. Explain your answer from the bias–variance trade-off perspective.

(5) (2 pts) Now assume the true model is

$$Y = \beta_1^* X_1 + \varepsilon,$$

i.e., $\beta_2^* = 0$. Would you expect the lasso estimator $\widehat{\boldsymbol{\beta}}^s$ (with optimally chosen s) to have smaller test MSE than that of the OLS estimator? Briefly explain.

Problem 2 (12 pts)

Let (Y_i, X_i) for $1 \leq i \leq n$ be training data with $X_i \in \mathbb{R}$ and $Y_i \in \mathbb{R}$. Consider the spline model with two knots $\xi_1 < \xi_2$:

$$\begin{aligned} Y_i &= f(X_i) + \varepsilon_i, \\ f(x) &= \beta_0 + \beta_1 x + \beta_2 (x - \xi_1)_+ + \beta_3 (x - \xi_2)_+, \end{aligned} \tag{2}$$

where $(x - \xi)_+ := \max\{0, x - \xi\}$. Assume $\mathbb{E}[\varepsilon_i] = 0$, $\mathrm{Var}(\varepsilon_i) = \sigma^2$, and ε_i's are independent. Throughout, take $\xi_1 = -0.5$ and $\xi_2 = 1$.

(1) (3 pts) Suppose

$$\beta_0 = 0, \quad \beta_1 = 1, \quad \beta_2 = -2, \quad \beta_3 = 1.$$

Sketch $f(x)$ as a function of x. Your sketch should clearly indicate the intercepts and slopes.

(2) (2 pts) Describe one method for estimating the coefficients $(\beta_0, \beta_1, \beta_2, \beta_3)$. Your answer should at least specify the optimization criterion used to obtain the estimates.

(3) (3 pts) After fitting the linear spline in Eq. (2), suppose we find underfitting and decide to fit a *natural cubic spline* with the same knots $\xi_1 = -0.5$ and $\xi_2 = 1$. Describe the constraints that define a natural cubic spline with these knots. Your answer should include the smoothness conditions imposed at each knot as well as the boundary (natural) conditions.

(4) (1 pt) Between the linear spline model and the natural cubic spline model (with the same knots), can you tell which model has smaller bias at a typical interior point $x \in (\xi_1, \xi_2)$? Briefly justify.

(5) (1 pt) Between the two models, can you tell which fitted model has larger variance? Briefly justify.

(6) (2 pts) Let the training MSE of a fitted model $\widehat{f}$ be

$$\mathrm{MSE}_{\mathrm{train}} = \frac{1}{n}\sum_{i=1}^{n}\big(Y_i - \widehat{f}(X_i)\big)^2$$

and let the expected MSE at a point x be

$$\mathrm{MSE}_{\mathrm{test}}(x) = \mathbb{E}\big[(Y - \widehat{f}(x))^2 \mid X = x\big],$$

where (X, Y) is an independent test draw from the same population.

(a) (1 pt) Which model, the linear or the natural cubic spline, do you expect to have smaller training MSE? Explain briefly.

(b) (1 pt) For the expected test MSE, can you conclude which model, the linear or the natural cubic spline, is smaller? Explain briefly.

Problem 3 (8 pts)

Cosider the same setting in **Problem 2** above.

(1) (3 pts) Fix knots $(\xi_1, \xi_2) = (-0.5, 1)$. Describe how to use 3-fold cross-validation to estimate the expected MSE of a fitted linear spline model. Your answer should clearly explain how the data are partitioned into folds, what model is fitted in each fold, how the prediction error is computed, and how the final cross-validation estimate is obtained.

(2) (2 pts) Suppose we want to select knots from

$$\xi_1 \in \{-2, -1, -0.5\}, \qquad \xi_2 \in \{0, 1, 3\},$$

with $\xi_1 < \xi_2$. Describe how to use cross-validation to choose the knot pair (ξ_1, ξ_2). (You may use 3-fold CV as in part (1).)

(3) (3 pts) Does repeating the cross-validation procedure in part (2) guarantee that the same knot pair is selected each time? If so, explain why. Otherwise, would increasing the number of folds help? Briefly justify your answer.

Problem 4 (10 pts, 1 pt for each subquestion)

Be sure to mark your answers on the **Scantron sheet**. There can be **one to four** correct answers to each question. One point is assigned to a multiple choice question **if and only if** all correct answers to this question are checked and no incorrect answer to this question is checked.

1. Which of the following statements about training error, test error, and overfitting are correct?

   A The test MSE is always larger than the training MSE.

   B An overfitted predictor is typically unstable with respect to small changes in the data.

   C Increasing the sample size can help reduce overfitting.

   D A smaller training error necessarily implies a smaller test error.

2. Which of the following statements about regression are true?

   A You can only fit a linear model when the true model is linear.

   B Multicollinearity leads the OLS estimator to have larger variance.

   C Adding more predictors to fit a new model can worsen prediction performance on test data.

   D One cannot use gradient descent to compute the OLS estimator.

3. Regarding regularization techniques in regression, which of the following statements are true?

   A Lasso regression can produce estimated coefficients that are exactly zero.

   B Ridge regression can be computed in a closed form.

   C Regularization reduces the bias but incurs extra variance.

   D Ridge regression tends to perform better than lasso when the true regression coefficients are sparse.

4. Which of the following statements about AIC/BIC and cross-validation are true?

   A AIC and BIC require specifying a likelihood for the model.

   B Cross-validation can be used for model selection without specifying a likelihood.

   C AIC and BIC can be applied only when the sample size is large.

   D Cross-validation can produce an estimate of the expected MSE.

5. Which of the following statements about gradient descent are true?

   A For convex optimization problems, gradient descent converges for any choice of step size.

   B It can only be used when the loss function is differentiable.

   C Using a smaller step size generally results in more stable but slower convergence.

   D Using a smaller step size always leads to fewer iterations until convergence.

6. In which case do we usually prefer the nonparametric method rather than the parametric method (the number of features is p and the sample size is n)?

   A When p is large and n is small.

   B When p is small and n is large.

   C When p is small and n is small.

   D None of the above statements is correct.

7. Which of the following statements are true?

   A Lasso can not be used when the number of features p is larger than the sample size n

   B Ridge regression can be applied even when p is larger than n.

   C Lasso regression generally does not have a closed-form solution.

   D Shrinkage regression is a parametric method.

8. Assume that the model $Y = f(X) + \varepsilon$ holds, where ε is a random noise with mean 0 and independent of X. Which of the following statements are true?

   A The regression function f minimizes the training MSE.

   B The regression function $f(x)$ minimizes the expected squared prediction error at $X = x$.

   C The expected squared prediction error of $f(x)$ is equal to $\mathrm{Var}(\varepsilon)$.

   D The regression function depends on the distribution of ε.

9. Which of the following statements are true?

   A Polynomial regression models may have bad behavior near the boundary of the feature space.

   B A fitted step function is piecewise constant and is typically discontinuous at the cut points.

   C Linear splines are piecewise linear and continuous.

   D Cubic splines are less complex than cubic polynomial regression.

10. Which of the following statements about local polynomial regression are true?

    A Local polynomial regression fits a polynomial model using only observations in a neighborhood of the target point.

    B Increasing the degree of the local polynomial always improves prediction performance.

    C The choice of neighborhood size (or bandwidth) affects the bias–variance trade-off.

    D Local polynomial regression can be applied when the number of features is large.

(You may use this page as scratch paper if needed)